

Contact Information

Mathematical Sciences Building, Office 609
Department of Mathematics, Purdue University

✉ epabonca@purdue.edu
🔗 ejpaboncancel.github.io

Research Interests

Dynamical Systems (e.g. Mathematical Physics, Symplectic Geometry, Modeling in Material Science), Machine Learning (e.g. Physics-Informed Neural Networks), Computational & Applied Mathematics (e.g. Numerical Methods, Mathematical Algorithms, Data Assimilation).

Education & Background

Doctor of Philosophy in Mathematics

May 2029

Purdue University, West Lafayette, Indiana | Advisor: Prof. Aaron Yip

Master of Science in Mathematics

May 2026

Purdue University, West Lafayette, Indiana

Bachelor of Science in Pure Mathematics (*Magna Cum Laude*)

December 2022

Curricular Sequence in Applied Mathematics for Science and Engineering

University of Puerto Rico, Mayagüez Campus (UPRM), Mayagüez, Puerto Rico | Mentor: Reyes M. Ortiz Albino

Skills and Other Information

Programming & Computation: Python, Julia, C++, SageMath, MATLAB | **Formatting & Tools:** HTML, $\text{\LaTeX}$, Git
Math & AI/ML: NumPy, SciPy, PyTorch, TensorFlow, JAX, Flux, CUDA | **Spoken Languages:** English and Spanish

Research Experience

URA-Sandia Graduate Student Summer Fellow

URA-Sandia Graduate Student Summer Fellowship (administered by ORISE and ORAU)

May 2025–August 2025

Computational & Information Sciences Foundation, Sandia National Laboratories

Livermore, California

Supervised by: Dr. Moe Khalil & Dr. Rileigh Bandy, Sandia National Laboratories

Research Topics: Machine Learning-based Surrogate Models, Data Assimilation

GEM Fellow Summer Research Program Intern

MIT Lincoln Laboratory Summer Research Program (GEM Fellowship Employer Sponsor)

May 2023–August 2023

Group 39, Division 3, MIT Lincoln Laboratory, Massachusetts Institute of Technology

Lexington, Massachusetts

Supervised by: Dr. Sam Polk & Dr. Mabel Ramírez, MIT Lincoln Laboratory

Research Topics: Classical Machine Learning, Bayesian Inference, Generative AI, Unsupervised Learning

Research Opportunity for Undergraduate Students in STEM

Puerto Rico Louis Stokes Alliance for Minority Participation

August 2019–December 2022

Department of Mathematical Sciences, University of Puerto Rico, Mayagüez Campus

Mayagüez, Puerto Rico

Supervised by: Prof. Reyes M. Ortiz Albino, University of Puerto Rico at Mayagüez

Research Topics: Number Theory, Commutative Algebra, Factorization Theory in Integral Domains

NSF REU Participant

- Summer@ICERM 2022: Computational Combinatorics

June 2022–August 2022

Institute for Computational and Experimental Research in Mathematics, Brown University

Providence, Rhode Island

Supervised by: Prof. Pamela E. Harris, University of Wisconsin-Milwaukee

Research Topics: Enumerative Combinatorics, Parking Functions

- NSF REU in Combinatorics, Probability and Algebraic Coding Theory

June 2021–August 2021

East Tennessee State University & University of Puerto Rico at Ponce

Johnson City, Tennessee

Supervised by: Prof. Fernando Piñero González, University of Puerto Rico at Ponce

Research Topics: Algebraic Geometry & Linear Error Correcting Codes

Projects

Project in AI Topic (TBD)

URA STEM Ambassadors Program

May 2026–December 2026

2026 URA STEM Ambassadors Challenge

TBD

Project in Biotechnology

MIT Lincoln Laboratory Summer Research Program

June 2023–July 2023

2023 MIT Lincoln Laboratory Intern Innovative Idea Challenge (I³C) | Mentors: Ryan Burrow and Ashok Kumar

SKINS: Skin-growth boosting and Intra-absorptive Solution bandages (3rd Place Project Winner)

Awards

Fellowships, Scholarships and Prizes

2025 Universities Research Association-Sandia National Labs Graduate Summer Fellowship

May 2025–August 2025

2023 National GEM Consortium PhD Science Fellowship

August 2023–May 2024

- Purdue University Department of Mathematics Sponsorship

August 2023–May 2024

- MIT Lincoln Laboratory Employer Sponsorship (Internship)

May 2023–August 2023

2023 MIT Lincoln Laboratory I³C 3rd Place Research Proposal Prize

July 2023

2022 Evertec Inc. STEM Scholarship

October 2022

Puerto Rico-Louis Stokes Alliance for Minority Participation Research Scholarship

August 2019–December 2022

Merits and Recognitions

2026 URA STEM Ambassador

March 2026

2023 Ford Foundation Predoctoral Fellowship Honorable Mention

March 2023

2022 Hispanic Scholarship Fund Scholar

June 2022

National Math Alliance Predoctoral Scholar

November 2021

UPRM Faculty of Arts and Sciences Honor Roll

August 2018–May 2023

Puerto Rico Department of Education Math Olympiad, 2nd Place Regional Winner

February 2018

National Trig-Star Math Competition, 16th Overall Finalist

June 2017

Eagle Scout Rank, with 2 Silver Palms

May 2017

Puerto Rico Department of Education Math Olympiad, 1st Place District Winner

February 2016, 2017

Papers and Articles (The asterisk symbol (*) denotes alphabetical order authorship.)

Research Articles and Preprints:

[1] S. Polk, E.J. Pabon-Cancel, R. Paleja, K. Chestnut-Chang, R. Jensen and M. Ramirez.

Unsupervised Behavior Inference from Human Action Sequences (UNBIAS).

2024 IEEE Conference on Games (CoG), Milan, Italy, 2024, pp. 1-8.

DOI: <https://doi.org/10.1109/CoG60054.2024.10645587>

[2] *A. Allen, E.J. Pabon-Cancel, F. Piñero-Gonzalez and L. Polanco.

Improving the Dimension Bound of Hermitian-Lifted Codes.

arXiv: <https://arxiv.org/abs/2302.01557>

[3] *D. Chen, P.E. Harris, J. Carlos Martinez Mori, E.J. Pabon-Cancel and G. Sargent.

Permutation Invariant Parking Assortments.

Enumerative Combinatorics and Applications, **4:1**, 1-25 (2024). #S2R4.

DOI: <https://doi.org/10.54550/ECA2024V4S1R4>

[4] *I. Byrne, N. Dodson, R. Lynch, E.J. Pabon-Cancel and F. Piñero-Gonzalez.

Improving the minimum distance bound of Trace Goppa codes.

Designs, Codes and Cryptography. **91**, 2649–2663 (2023).

DOI: <https://doi.org/10.1007/s10623-023-01216-6>

Contributions to the profession:

[1] *P.E. Harris, Z. Markman, L. Martinez, A. Mock, E.J. Pabon-Cancel, A. Verga, and S. Wang.

A Model for a One-Hour Workshop on Mentoring.

MAA Focus, **43**(1), 18-21 (2023).

Service and Outreach

Academic Service

- Universities Research Association

URA STEM Ambassadors Program

University Ambassador (Purdue University)

March 2026–present

West Lafayette, Indiana

Teaching and Grading

- MA 26100: Multivariate Calculus (Asynchronous)

June 2026–August 2026

- MA 59500MB: Mathematical Biology (Grading) January 2026–May 2026
- MA 32500: History of Mathematics (Grading) January 2026–May 2026
- MA 26100 REC: Multivariate Calculus Recitation (Teaching) August 2025–December 2025
January 2025–May 2025
August 2024–December 2024
- MA 13900: Mathematics for Elementary Teachers III (Grading) June 2024–August 2024

Student Associations

- Purdue SACNAS Chapter
Purdue University
Member August 2025–May 2026
West Lafayette, Indiana
- PythagORUM
University of Puerto Rico, Mayagüez Campus
Co-founder & Vice-President August 2022–December 2022
Mayagüez, Puerto Rico
- Society of Physics Students, UPRM Chapter
University of Puerto Rico, Mayagüez Campus
Committee Assistant August 2018–December 2022
Mayagüez, Puerto Rico

Poster Sessions, Panels, Presentations and Conferences

- Purdue University Student Computational and Applied Mathematics Seminar
Helen B. Schleman Hall, Purdue University 27 March 2026
Panel: Internships and Job Opportunities Beyond Academia West Lafayette, Indiana
- Purdue Graduate Student Government Research-O-Rama
West Lafayette Public Library 21 February 2026
Poster: Data-Driven Closure Models (DDCMs) West Lafayette, Indiana
- Purdue University Prospective Student Visit Weekend Graduate Student Research Expo.
Mathematical Sciences Building, Purdue University 20 February 2026
Poster: Data-Driven Closure Models (DDCMs) West Lafayette, Indiana
- Purdue University Student Computational and Applied Mathematics Seminar
Helen B. Schleman Hall, Purdue University 13 February 2026
Presentation: Data-Driven Closure Models (DDCMs) West Lafayette, Indiana
- URA-Sandia Graduate Summer Fellowship Lightning Talk
Presentation: Data-Driven Closure Models (DDCMs) 6 August 2025
Virtual Seminar
- Sandia National Laboratories CA SIP Intern Symposium
Auditorium, Sandia National Laboratories-Livermore 5 August 2025
Poster: Data-Driven Closure Models (DDCMs) Livermore, California
- Combinatorics and Coding Theory in the Tropics (UPR-Ponce)
Invited REU Seminar: My Story & Permutation-Invariant Parking Assortments 18 July 2025
Virtual Seminar
- Purdue University Student Commutative Algebra Seminar
Helen B. Schleman Hall, Purdue University 18 November 2024
Presentation: Results in $\tau_{(n)}$ -factorizations and $\tau_{(n)}$ -primes. West Lafayette, Indiana
- Purdue University Student Math History Seminar
Lawson Computer Science Building, Purdue University 9 September 2024
Presentation: Testimonios: Stories of Latinos and Hispanics in Mathematics West Lafayette, Indiana
- Underrepresented Students in Topology and Algebra Research Symposium 2024
University of Iowa 20-21 April 2024
Iowa City, Iowa
- 2023 MIT Lincoln Lab Intern Innovative Idea Challenge
MIT Lincoln Laboratory Auditorium 14, 21 July 2023
Poster: Skin-Absorptive and Skin-Growth Boosting Bandages Lexington, Massachusetts
Presentation: SKINS: Skin-growth boosting and Intra-absorptive Solution Bandages

- Combinatorics and Coding Theory in the Tropics (UPR-Ponce)
Invited REU Seminar Talk: Graduate School: Application tips and advice 7 July 2023
Virtual Seminar
- 2023 ACS Junior Technical Meeting-Puerto Rico Interdisciplinary Scientific Meeting 29 April 2023
University of Puerto Rico at Bayamón, Sponsored by PR-LSAMP Bayamón, Puerto Rico
Presentation: Properties of $\tau_{(n)}$ -primes
- 38th Interuniversity Mathematical Sciences Research Seminar 24-25 February 2023
University of Puerto Rico, Mayagüez Campus Mayagüez, Puerto Rico
Presentation: Permutation Invariant Parking Assortments
- 2023 AAAS Emerging Researchers National Conference in STEM 9-11 February 2023
Omni Shoreham Hotel Washington, District of Columbia
Poster: Permutation Invariant Parking Functions with cars of assorted lengths
- Joint Mathematics Meetings 2023 4-7 January 2023
John B. Hynes Veterans Memorial Convention Center Boston, Massachusetts
Poster: Permutation Invariant Parking Functions with cars of assorted lengths
Presentation: Permutation Invariant Parking Functions with Cars of Arbitrary Lengths
- Field of Dreams Conference 2022 4-6 November 2022
The Graduate Hotel, University of Minnesota-Twin Cities Minneapolis, Minnesota
- 2022 SACNAS National Diversity in STEM Conference 27-29 October 2022
Pedro Roselló Convention Center San Juan, Puerto Rico
Poster: The Study of $\tau_{(n)}$ -primes
- 2022 Gulf Coast Undergraduate Research Symposium 8-9 October 2022
William Marsh Rice University Houston, Texas
Presentation: Properties of $\tau_{(n)}$ -primes
- Summer@ICERM 2022: Computational Combinatorics 3 August 2022
Institute for Computational and Experimental Research in Mathematics Providence, Rhode Island
Presentation: On Permutation-Invariant Parking Sequences
- 2022 ACS Junior Technical Meeting-Puerto Rico Interdisciplinary Scientific Meeting 9 April 2022
University of Puerto Rico at Humacao Humacao, Puerto Rico
Presentation: The Study of $\tau_{(n)}$ -primes
- Joint Mathematics Meetings 2022 6-9 April 2022
Poster: Improving Bounds of Hermitian-Lifted Codes Virtual Conference
- 37th Interuniversity Mathematical Sciences Research Seminar 25-26 February 2022
Poster: The Study of $\tau_{(n)}$ -primes Virtual Conference
Presentation: Improving Bounds of Hermitian-Lifted Codes
- 2021 Math REU Conference@Clemson University 19 July 2021
Presentation: Improved Hermitian-Lifted Codes Virtual Conference
- 2021 ACS Junior Technical Meeting-Puerto Rico Interdisciplinary Scientific Meeting 23-24 April 2021
Presentation: The Study of $\tau_{(n)}$ -atoms Virtual Conference
- 35th Interuniversity Mathematical Sciences Research Seminar 6-7 March 2020
University of Puerto Rico at Cayey Cayey, Puerto Rico
Poster: The Study of $\tau_{(n)}$ -atoms

Academics and Coursework

Purdue University

Coursework (Graduate):

MA 61500: Numerical Methods for Partial Differential Equations

January 2026–May 2026

MA 53200: Elements of Stochastic Processes

January 2026–May 2026

MA 59800ZAT: Hamiltonian Dynamics

January 2026–May 2026

MA 55400: Linear Algebra

August 2025–December 2025

MA 51900: Introduction to Probability	August 2025–December 2025
MA 57300: Numerical Solutions of Ordinary Differential Equations	August 2025–December 2025
MA 59500OT: Computational Optimal Transport and Deep Generative Models	January 2025–May 2025
MA 59800ZY: Topics in Dynamical Systems (Bifurcation Theory)	January 2025–May 2025
MA 54600: Introduction to Functional Analysis	January 2025–May 2025
MA 59500AFF: Analytic Theory of Function Fields	August 2024–December 2024
MA 59500MM: Introduction to Mathematical Modeling	August 2024–December 2024
MA 54300: Introduction to Ordinary Differential Equations and Dynamical Systems	January 2024–May 2024
MA 54400: Real Analysis and Measure Theory	January 2024–May 2024
MA 55300: Introduction to Abstract Algebra	August 2023–December 2023
MA 53000: Functions of a Complex Variable I	August 2023–December 2023

University of Puerto Rico, Mayagüez Campus

Coursework (Graduate):

MATE 6101: Number Theory I	August 2022–December 2022
MATE 5150: Linear Algebra	January 2021–May 2021

Coursework (Undergraduate):

MATE 4000: Elements of Topology	August 2022–December 2022
MATE 4052: Advanced Calculus II	January 2022–May 2022
MATE 4051: Advanced Calculus I	August 2021–December 2021
MATE 4061: Numerical Analysis I	August 2021–December 2021
MATE 4021: Introduction to Mathematical Logic	August 2021–December 2021
MATE 4010: Introduction to Complex Variables	January 2021–May 2021
ESMA 4001: Mathematical Statistics I	January 2021–May 2021
MATE 4020: Partial Differential Equations and Fourier Series	August 2020–December 2020
MATE 4008: Introduction to Algebraic Structures	August 2020–December 2020
MATE 3040: Theory of Numbers	August 2020–December 2020
MATE 4009: Ordinary Differential Equations	June 2020–July 2020
MATE 4031: Introduction to Linear Algebra	January 2020–May 2020
COMP 3010: Introduction to Computer Programming I	January 2020–May 2020
MATE 3063: Calculus III	January 2020–May 2020
MATE 3032: Calculus II	August 2019–December 2019
MATE 3020: Introduction to Mathematical Foundations	August 2019–December 2019
MATE 3031: Calculus I	June 2019–July 2019
MATE 3172: Precalculus II	January 2019–May 2019
MATE 3171: Precalculus I	August 2018–December 2018